II. Give very short answer to the following questions.

21. Define regression.
22. What are the types of regression?
23. Write the two simple linear regression equations.
24. Write the two simple linear regression coefficients.
25. If the regression coefficient of X on Y is 16 and the regression coefficient of Y on X is 4, then find the correlation coefficient.
26. Find the standard deviation of Y given that $V(X)$ is 36, $b_{XY} = 0.8$, $r_{XY} = 0.5$.

III. Give short answer to the following questions.

27. Define simple linear and multiple linear regressions
28. Distinguish between linear and non-linear regression.
29. Write the regression equation of X on Y and its normal equations.
30. Write the regression equation of Y on X and its normal equations.
31. Write any three properties of regression.
32. Write any three uses of regression.
33. Write any three differences between correlation and regression.
34. If the regression equations are $\hat{X} = 64 - 0.95y$, $\hat{Y} = 7.25 - 0.95x$ then find the correlation coefficient.
35. Given the following lines of regression.

    $8X - 10Y + 66 = 0$ and $40X - 18Y = 214$. Find the mean values of X and Y.

36. Given $\overline{x} = 90$, $\overline{y} = 70$, $b_{XY} = 1.36$, $b_{YX} = 0.61$ when $y = 50$, Find the most probable value of X.
37. Compute the two regression equations from the following data.

x	1	2	3	4	5
y	3	4	5	6	7

    If $x = 3.5$ what will be the value of $\hat{Y}$?

IV Give detailed answer to the following questions.

38. Write in detail the properties of regression.
39. Explain in detail the uses of regression analysis.
40. Distinguish between correlation and regression.
41. Interpret the result for the given information. A simple regression line is fitted for a data set and its intercept and slope respectively are 2 and 3. Construct the linear regression of the form $Y = a + bx$ and offer your interpretation for 'a' and 'b'. If X is increased from 1 to 2, what is the increase in Y value. Further if X is increased from 2 to 5 what would be the increase in Y. Demonstrate your answer mathematically.

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